HT-based Model Equality Testing

Preliminary Progress Report

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Introduction

Motivation:

- LLMs are core foundation models for many AI applications.
- Training them from scratch is extremely costly; misuse of released weights can pose commercial and ethical risks.
- Many LLMs are closed-source or only conditionally open under restrictive licenses or approval processes.

$\Rightarrow$ Identifying the relationships between models is crucial.

Problem: Existing methods rely on outputs or weight alignment, but these are inaccessible or fragile in many scenarios.

This Work: Use spectral shape ($\texttt{PL_}\alpha$) as a lightweight fingerprint to compare models directly from weights.

Research Question

Core Question

- Can layerwise spectral shape (e.g., power-law exponents) provide a lightweight fingerprint of structural information?
- Can this fingerprint distinguish related vs. unrelated LLMs, especially when outputs or activations are inaccessible?

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Prior Works

Two directions in recent literature:

- **Output-based Identifier:** Use generated texts to identify source models/Test whether outputs of two models are statistically identical
  - *Idiosyncrasies in LLMs (2025)*
  - *Model Equality Testing (ICLR 2025)*
- **Structure-aware independence tests:** Use weights/activations to infer training relationship
  - *Independence Tests for LMs (ICML 2024)*
  - *HuRef: HUman-REadable Fingerprint for Large Language Models (NeurIPS 2023)*

Prior Work: Output-Based Attribution

Method: Identify LLMs based on their output styles and semantics.

Example: *Idiosyncrasies in LLMs (2025)*

- Trains classifiers to predict source model from text samples.
- Captures model-specific lexical and formatting bias.
- Robust to paraphrasing, prompt variation, and sampling randomness.
- Requires access to model outputs; unsuitable for structural analysis.

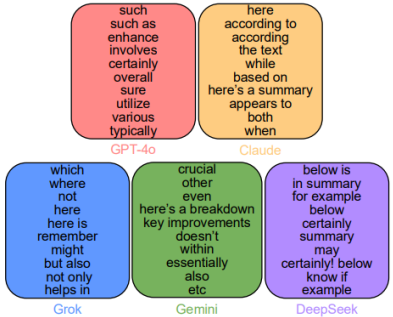


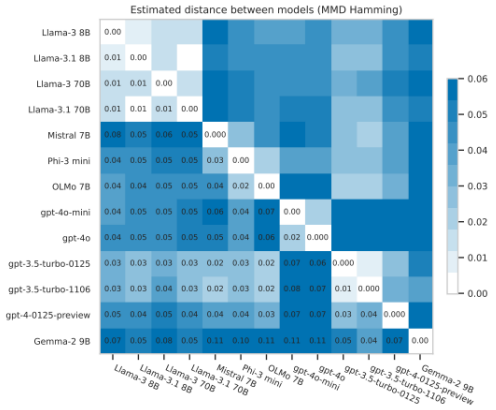
Figure 6. **Characteristic phrases.** We train a logistic regression model on TF-IDF features of chat APIs' outputs and extract the top 10 phrases for each LLM based on the coefficients of these features. We remove common words shared across these LLMs.

Prior Work: Output-Based Statistical Testing

Method: Test model equivalence by comparing outputs on shared inputs.

Example: *Model Equality Testing (ICLR 2025)*

- Treat models as black boxes; compute similarity over output distributions.
- Uses distance metrics (e.g., Wasserstein, PL_based) on embeddings.
- Formulates hypothesis test to decide if models are statistically identical.
- Output-dependent; fails under obfuscation or stochastic decoding.



Prior Work: Structure-Aware Independence Tests

Method: Use internal weights/activations to assess model similarity.

Example: *HuRef: HUman-REadable Fingerprint for Large Language Models (NeurIPS 2023)*

- Flatten all weight matrices and biases of an LLM into vectors and concatenate them together into a single huge vector.
- Derives invariant terms to cope with weight rearrangements
- Computes cosine similarities.
- Maps the invariant terms to image with imageGAN.

Model	Alpaca-Lora	Alpaca	Chinese-LLaMA	Vicuna	Baize	MedAlpaca	Koala	WizardLM	MiniGPT-4	Chinese-Alpaca	Baichuan	OpenLLaMA	InternLM	LLaMA-2
PCS	99.87	99.91	99.68	99.80	99.73	99.90	99.82	99.89	99.70	99.52	0.83	1.16	0.28	1.51
ICS	99.60	99.95	93.57	99.42	99.60	99.86	99.63	99.89	99.20	91.35	0.32	0.32	0.03	3.16

Table 1: The cosine similarities of model parameters (PCS) and invariant terms (ICS) between various LLMs w.r.t. the LLaMA-7B base model. All models are of the same size.

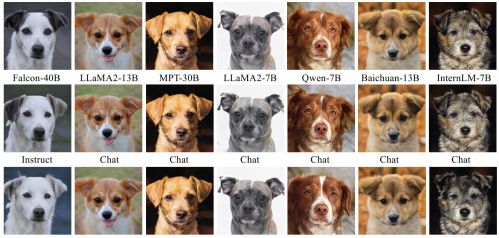


Figure 5: Fingerprints of 7 different base models (in the first row) and their corresponding offspring models (the lower two rows) are presented. The base model's name is omitted in the offspring models.

Prior Work: Structure-Aware Independence Tests

Method: Use internal weights/activations to assess model similarity.

Example: *Independence Tests for LMs (ICML 2024)*

- Leverages permutation symmetry in MLPs to define null distributions.
- Matches hidden units using activation similarity (ϕ_{MATCH}).
- Computes exact p-values via permutation tests.
- Requires access to internal states; unsuitable for black-box models.

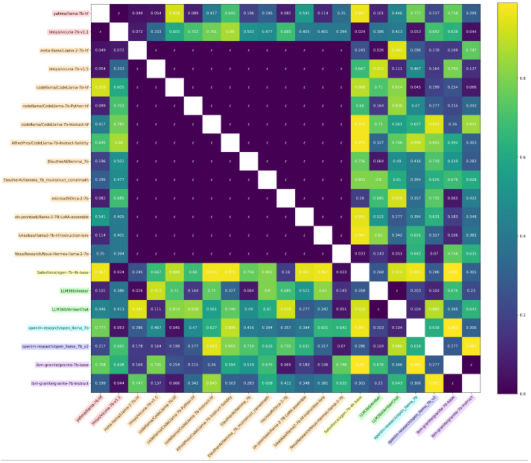


Figure 9: We report values of ϕ_{MATCH} on Llama-7B model pairs, where $\varepsilon = 2.2\text{e-}308$. We use `MATCH` on ϕ_{MATCH} and aggregate with `FISHER` across the 32 MLPs.

Heavy Tail Theory Spectral Fingerprint

Background: HT-SR Theory & RMT

As training proceeds, the Empirical Spectral Density follows ***random → structured*** *state.*

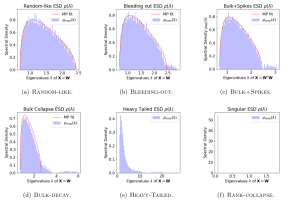
The ESD of the weight covariance $X = W^\top W$ evolves from Marcenko Pastur (MP) distribution ⇒ MP with outliers ⇒ heavy-tailed.

The tail often follows a power-law: $P(\lambda) \sim \lambda^{-\alpha}$.

Interpretation:

- Smaller α means heavier tail ⇒ stronger learned structure.
- ESD metrics are interpretable and validated

So layerwise α might serve as a lightweight fingerprint to compare models.



(a) Random-like. (b) Bleeding-out. (c) Bulk+Spikes.
(d) Bulk-decay. (e) Heavy-Tailed. (f) Rank-collapse.

Figure 1: Evolution of ESD

Research Gap and Design Motivation

Limitations of prior work:

- **Output-based methods**
  - fail when models are API-only or obfuscated
  - sensitive to sampling and input choice
- **Structure-aware methods**
  - require alignment
  - computationally expensive

Positioning this Work: ESD metrics like `Power-law_`α values capture meaningful structural information.

- Interpretable
- Lightweight to extract
- Invariant to most reparametrizations.

Goal: Use layerwise spectral patterns to reflect the training process, and detect whether two models are structurally equivalent or derived from one another.

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Pipeline Overview

Goal: Compare LLMs using layerwise spectral shape metrics (e.g., PL_α).

Step 1: Metric Extraction

- For each model, extract spectral metrics (e.g. α, λ, $\mu_{ks_distance}$) of the weight/activation matrix from each module in each layer.
- Supports MLP, attention, activation modules.
- for a LLM with m modules and l layers, its fingerprint $A \in \mathbb{R}^{l \times m}$, where the $i \times j$-th entry represents the spectral metric of the j-th module in the i-th layer.

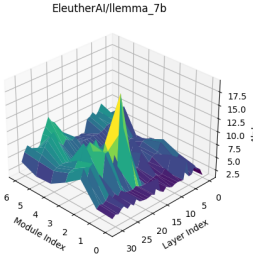


Figure 2: Fingerprint

Step 2: Structure-Aware Comparison

- **Same architecture:** Calculate the spearman correlation of each module between two LLMs, then derive the average score.
- **Different architecture:** Use LAP to match layers by spectral similarity.

Spectral Fingerprint: Visualization Example

To illustrate the spectral structure, we extract PL_α values across all modules and layers of two unrelated models: **Llemma-7B** & **Open-Llama-7B**

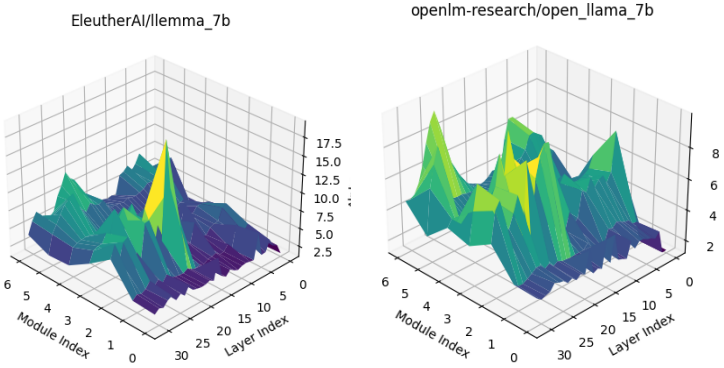


Figure 3: 3D surface plots of PL_α values per layer and module.

Model Selection: Focus of Current Experiments

Model Pair Type	**Example**	**Purpose**
Same base, diff objective	LLaMA-7B vs Vicuna-7B	Impact of fine-tuning task
Base vs adapter tuning	LLaMA-7B vs LoRA-tuned model	Effect of structural injection
Diff size, same family	LLaMA-7B vs LLaMA-13B	Detect structural scale shifts
Cross-family architecture	LLaMA-7B vs GPT-2 XL / Gemma	Generalizability of fingerprinting

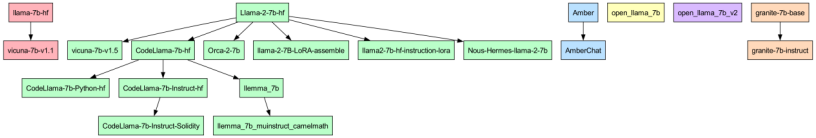


Figure 4: Model Families

Primary Results: Averaged

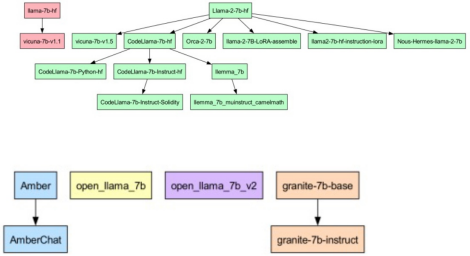


Figure 5: Model Family

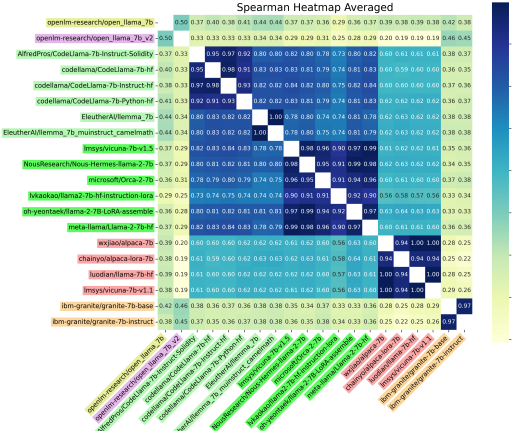


Figure 6: Averaged Spearman Correlation

Primary Results: MLP

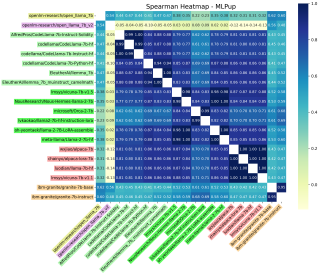


Figure 7: MLP Up

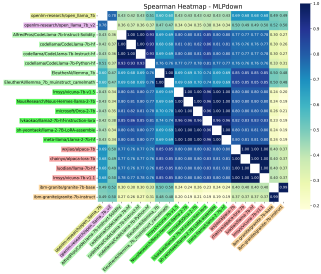


Figure 8: MLP Down

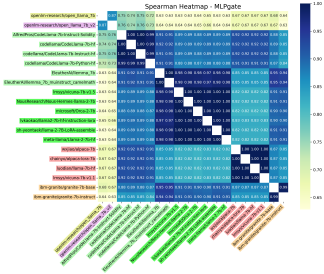


Figure 9: MLP Gate

Primary Results: Attention

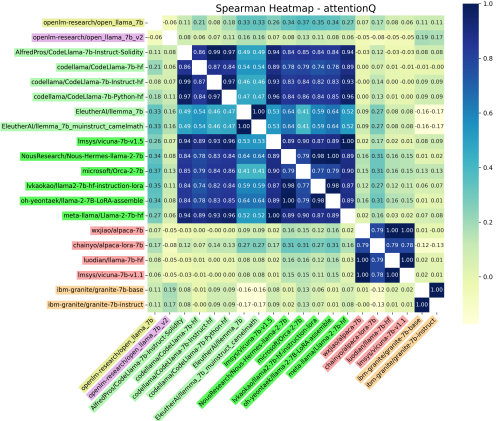


Figure 10: Attention Query

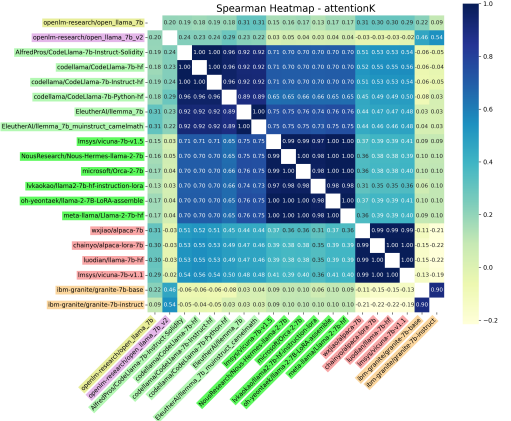


Figure 11: Attention Key

Primary Results: Attention

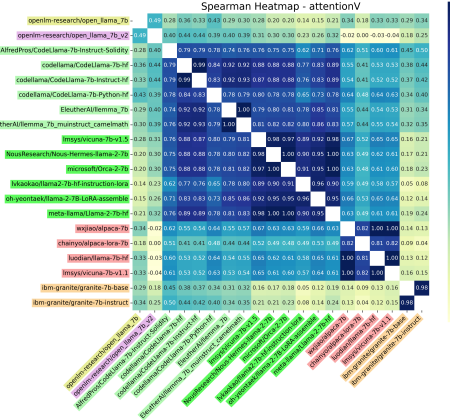


Figure 12: Attention Value

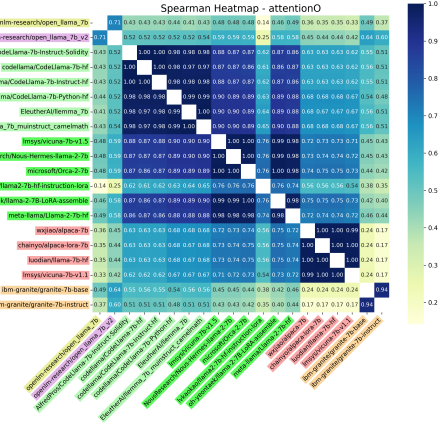


Figure 13: Attention Output

Key Observations: Module Sensitivity & Fingerprint Strategy

Observation:

1. Different modules encode different aspects.
   MLP-down and gate distinguish task-specific subfamilies; MLP-up is less reliable across families.
2. Some fine-tuning techniques like LoRA may obfuscate the metric of specific modules.

Strategy:

1. **Module selection:**
   - Focus on Q/K for base similarity.
   - Use MLP-down/gate for task separation.
   - Skip modules like MLP-up if consistently noisy.
2. **Fingerprinting validation:**
   Build a library of typical α-vectors for known families. Compare new models to the library to infer possible lineage.

Key Observations: Randomness vs. Metric Simplicity Trade-off

Observation:

- Weight-level independence tests leverage the irreproducible randomness in model weights.
- α-based spectral fingerprint doesn't require any fine alignment, but it compresses the structure and drops most noise detail. So it could be blind to the subtle independence signal that other methods capture.

Example from 4 toy models: A and C are trained on the same dataset independently. D is trained on another dataset. B is a fine-tuned version of A.

Current Direction: Consider smaller singular values to recover training differences.

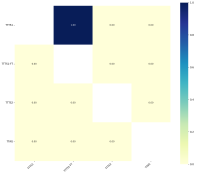


Figure 14: Cosine

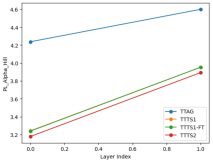


Figure 15: Alpha

Key Observations: Metric Sensitivity

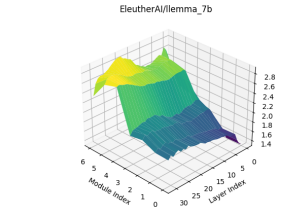
Observation:

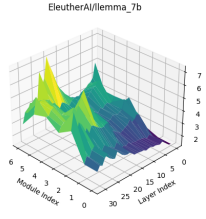
1. Task-specific fine-tuning often shifts only a few large singular values.
2. Different fitting strategies contributes to different metric shapes.

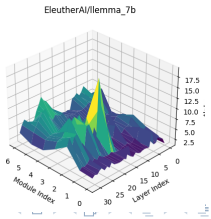
Risk:

1. If α is fully determined by a few large directions, it may overreact to minor changes.
2. Inconsistent or badly chosen fit regions could yield misleading fingerprints.

Current Direction: Determine the best fitting strategy and run sufficient experiments to see if problem exist







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Step 1: Toy Baselines for Fingerprint Validation

1. Random baseline

- Generate toy models with random initialization.
- Compute pairwise Spearman Rank.
- Confirm SR ≈ uniform for unrelated random weights.
- Define possible threshold for noise floor.

2. LoRA false negative check

- Test LoRA-tuned models with known modules (e.g., V/Q).
- Check if local α drops produce false negatives.
- Verify if average SR still holds up.

Step 2: Cross-Architecture Matching Risk

Key concern: Different architectures ⇒ layer counts, modules may differ.

- Layer matching (e.g., Hungarian algorithm) pairs layers by similarity.
- After matching, using SR on matched pairs may inflate unrelated models.

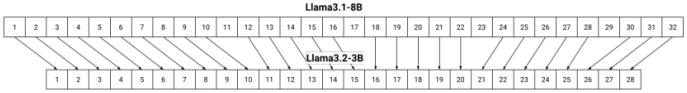


Figure 4: We evaluate $\phi^{(i,j)}_{\text{MATCH}}$, the unconstrained setting statistic, between all pairs of GLU MLPs in Transformer block $i \in \{1, 2, \ldots, 32\}$ of Llama 3.1-8B and Transformer block $j \in \{1, 2, \ldots, 28\}$ of Llama 3.2-3B. Arrows indicate if $\phi^{(i,j)}_{\text{MATCH}} <$ 1e-4 and suggest which Transformer blocks of Llama 3.1-8B were kept in the pruning process to initialize Llama 3.2-3B.

Test plan: Compare pre-match vs. post-match SR behavior

- Same-arch shuffle: match shuffled vs. original layers, check if SR produces false positives.
- Cross-arch test: LLaMA vs. GPT/Yi with LAP matching.

⇒If matched SR inflates noise, we may need an extra filter: e.g., KS distance, L2.

Extension: Identifying Crucial Singular Values

Definition:

For singular vector v_i and activation direction f_j:

$$\text{Overlap}(v_i) = \max_j |\langle v_i, f_j \rangle|$$

Proposed Use:

- Extract top-N small singular vectors for selected modules.
- Compute overlap sequence per model.
- Use Spearman Rank of these sequences as fine-grained fingerprint.

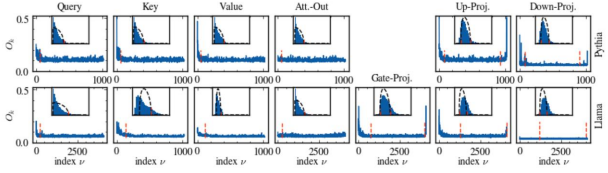


Figure 4. Overlaps in block 10 of Pythia and Llama. All matrices directly involved in the attention mechanism (Query, Key, Value) exhibit larger singular values outside the MP spectrum, with corresponding singular vectors showing high overlap with the activation covariance matrix. Attention-Output matrices remain within the MP distribution (no significant outliers or overlaps). For Pythia, Up- and Down-Projection matrices have outliers at both the small- and large-value ends, correlated with increased overlap. In Llama, only the Gate- and Up-Projection matrices display such outliers and strong overlap; the Down-Projection matrix does not show small outliers. For clarity, the MP distribution is drawn using the empirical variance of each matrix.

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Thank you for listening !

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